

TIC 403030272

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 9.455345 d (ALIASED, see below)
-> corrected period: 18.910690 d
Epoch (t0) 2368.2066 BTJD
Duration 0.2440 d
Depth 11.702 % (117,021 ppm)
SNR 518.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

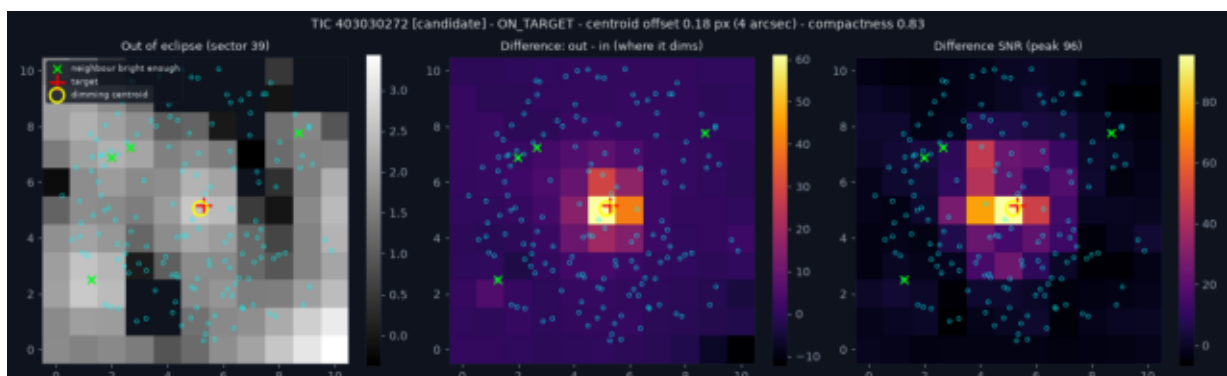
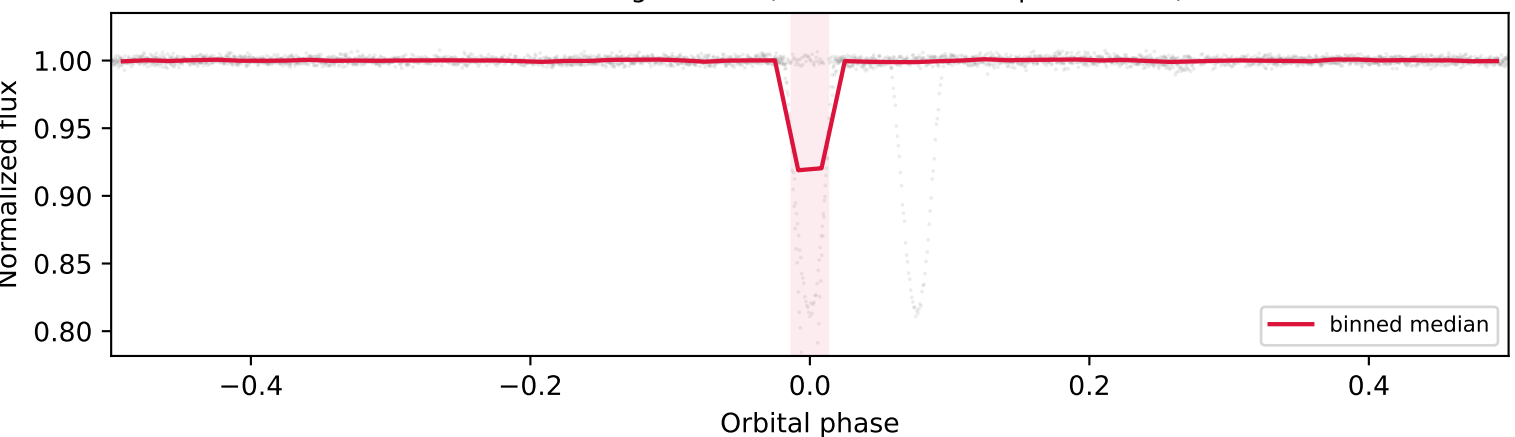
Eclipses observed 3 (gate ≥ 3)
Robust depth ratio 1.06 (gate ≥ 0.50)
Robust depth sigma 440.9 (gate ≥ 7.0)
Odd/even z-score 47.40 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 39 light-curve dip in this sector: 499.7 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 95.69 (gate ≥ 5.0) Compactness 0.83 (gate ≥ 0.50)
Dimming centroid offset 0.18 px (4 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 403030244
3.34 px away, dTmag +1.60 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 47.40 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 18.910690 d, in any write-up.